Coursework 2 – Functions, Arrays and Pointers

Instructions: For the following exercises, you will be writing programmes in a **single** DevC++ file, which you are going to submit on BlackBoard with the following filename formal: **"Your name" – CW2**. Use the appropriate comments, techniques and printed messages so that programmes are prompting and guiding the user. **DO NOT** copy from each other. **Similar works will be awarded zero marks. DO NOT** attempt to use AI assistants. Failure to adhere to the instructions will result in loss of points (-1 for filename, -5 for non-single file submissions, -1 for badly written and unreadable code)

1. Write a programme in C which:

   a. will ask the user to enter 10 integer numbers and then storing them in an **one-dimension array**

   b. then a function **void minmax(int arg[], int size)** will detect and print the minimum and maximum values of the array

   c. then ask the user whether he wants to the programme to print the sum of all elements in the array. The sum will be calculated by the function **int sumel(int arg[], int size)**

2. For the evaluation of the value of a polynomial, write a C – programme that:

   a. will ask the user to determine the degree of the polynomial

   b. then fill an **one-dimension array** of **real** numbers with the coefficients of the polynomial, starting from the **highest-ordered** term and ending with the constant. Then, the function **void poly(double arg[], int deg)** will print the polynomial in descending powers of x
   (Ex. $\boldsymbol{f(x) = a_n x^n + a_{n-1} x\text{^}(n-1) + \ \ldots + a_0}$)

   c. then write the function **double eval(double arg[], int deg, double x)** which will evaluate the polynomial of degree "deg", whose coefficients are stored in the "arg[]", by substituting the value of "x"

3. The executives of an electrical installations company is constructing 20 photovoltaic (PV) farms all over the country and they hire you to write a C programme with the following specifications:

   a. The programme should be **easily upgradable**, in case more PV farms are to be constructed

   b. Ask the user to register the daily production in MW from each farm in an **one-dimension array** of **integers**, named "production"

c. Then **sort** and **print** the "production" array in ascending order. This is done by writing a function, **void bsort(int arg[], int size)**, which it will take an **array** and its **size** as the arguments and execute the **bubble-sort** algorithm. Also, write another function **void swap(int *a, int *b)** to assist with the exchange between two elements of the array during the bubble-sort. For the calling of this function remember how **scanf()** is syntaxed

d. Next print the sum of the 5 highest production entries and label them as "Industry total", then the sum of the 5 lowest production entries and label them as the "Rural total" and the sum of the rest to be labelled as "Cities total"